



In today's time, safety and comfort are paramount, and automation is playing an active role in enhancing lifestyle, safety, and comfort along with improved efficiency. And to realise automation, gathering and processing data is the most crucial part, which is taken care of by the Internet of Things (IoT).

IoT is simple yet complex

ered device performing a simple task, which can easily be performed by even a low-cost smartphone, or a decade-old microcontroller (MCU) combined with a simple sensor that is easily available in the market.

The real challenge here is not to complete the task or gather data but to do all of it in a space-constrained, budget-constrained, power-constrained system that can work for years without any human intervention. And not to forget about the security, which in an IoT system is paramount. Furthermore, there is a time-relevance aspect to the IoT solution, that is, the product and its components should remain relevant for at least 5 to 10 years.

“The IoT solution has at least a 5-to-10-year life,” says Manish. “So, there is a time relevance aspect of the hardware and software, both of which cannot go obsolete. You must be able to upgrade the capability of your devices, capabilities of your hard-